



Survey paper

FIDWATCH: Federated incremental distillation for continuous monitoring of IoT security threats

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ABSTRACT

The fast evolutions of Internet of Things (IoT) technologies have been accelerating their applicability in different sectors of life and becoming a pillar for sustainable development. However, this revolutionary expansion led to a substantial increase in attack surface, raising many concerns about security threats and their possible consequences. Machine learning has significantly contributed to designing intrusion detection systems (IDS) but suffers from critical limitations such as data privacy and sovereignty, data imbalance, concept drift, and catastrophic forgetting. This collectively makes existing IDSs an improper choice for securing IoT environments. This paper presents a federated learning approach called FIDWATCH to continuously monitor and detect a broad range of IoT security threats. The local side of FIDWATCH introduces contrastive focal loss to enhance the ability of the local model (teacher) to discriminate between diverse types of IoT security threats while putting an increased emphasis on hard-to-classify samples. A fine-grained Knowledge Distillation (KD) is introduced to allow the client to distill the required teacher's knowledge into a lighter, more compact model termed the pupil model. This greatly assists the competence and flexibility of the model in resource-constrained scenarios. Furthermore, an adaptive incremental updating method is introduced in FIDWATCH to allow the global model to exploit the distilled knowledge and refine the shared dataset. This helps generate global anchors for improving the robustness of the mode against the distributional shift, thereby improving model alignment and compliance with the dynamics of IoT security threats. Proof-of-concept simulations are performed on data from two public datasets (BoT-IoT and ToN-IoT), demonstrating the superiority of FIDWATCH over cutting-edge performance with an average f1-score of 97.07% and 95.63%, respectively.

1. Introduction

With the lightning-fast pace by which the Internet of Things (IoT) technology is advancing, it is now recognized as an essential factor in the progression of a wide range of application areas, including smart homes, transportation, healthcare, and education. As IoT technology advances, the security requirements for devices, systems, and applications become increasingly stringent for academic institutions and commercial enterprises. It is projected that over 50 billion IoT devices will be in use globally by 2030 [1]. However, utilizing conventional security methods to safeguard IoT devices is not feasible because they function

autonomously, have scarce resources, and are online. Major threats to the availability, privacy, and security of IoT devices and networks include physical attacks, network intrusions, and data breaches. If an attacker can breach an IoT device, they can use it to execute attacks. There are many advantages to IoT, but a significant downside is the exposure of hundreds of millions of unprotected devices to various security threats. A high number of nearly unprotected, networked devices is a major security risk in IoT setups. [2].

Deep Learning (DL) has demonstrated a significant potential for identifying and protecting against attacks and viruses, and it has revolutionized the implementation of IoT security. Conventional DL

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researches rely on domain expertise and feature extraction to build a set of attributes for learning. It became progressively challenging to manually upgrade the training elements to keep up with the ever-evolving security threats. This is because a continuous stream of new security threats is created, and existing strains are upgraded with new and different features [3]. Stationary, variable, or composite structures are typically used for intrusion detection. Nevertheless, there are several cases where this is not true. High-level static themes or dynamic behavioral norms are typically used in intrusion detection for network devices. Static methods, however, are unsuitable for IoT ecosystems since they rely on analyzing assembly code and file structure [4]. This is because the CPUs in IoT devices always have varying designs. However, dynamic analysis can only examine an attack's actions along one particular path. For this reason, traversing all possible paths is challenging, necessitating a high degree of running time and high computing requirements for this class of procedures [5-7].

Unfortunately, the common DL methods for intrusion detection usually encounter many challenges that limit their applicability in the real world. They relied on centralized/distributed training using stationary datasets that might encounter data imbalance and concept drift over time. The data imbalance is caused by the underrepresentation of certain classes of attacks, which might make the model biased toward certain types of attacks and brawls to identify sporadic or unknown threats. On the other hand, concept drift represents a scenario in which the statistical characteristics of data vary across IoT devices, reducing the attack detection performance [8-10]. Moreover, DL-based solutions suffer from catastrophic forgetting, resulting in networks losing earlier knowledge learned after being exposed to new IoT traffic. This can be attributed to the iterative nature of training a DL model. Gradient updates rely only on the most recent data samples, which implies progressive loss/forgetting of earlier representations in the learning process [1],[2],[11]. Therefore, it is necessary to research and investigate how to develop a novel applicable FL solution for intrusion detection that can avoid the aforementioned problems while keeping high generalization performance. In this work, FIDWATCH is proposed as a new Federated Learning (FL) framework personalized explicitly to continuously and efficiently detect security threats in the dynamics of the IoT world.

The main contributions of the paper are summarized as follows. First, to enforce the local model (acting as a teacher) to focus on hard-to-classify examples, we proposed a contrastive focal loss to take advantage of contrastive learning and focal loss. This is achieved through arbitrary masking of traffic flows, which enables the model to discriminate precisely between different classes of IoT security threats. Second, a fine-grained KD approach is introduced based on combinatorial loss that enables the distillation of knowledge from the teacher to a pupil model, facilitating resource efficiency. Third, FIDWATCH incorporates an incremental updating mechanism that enables continuous model

improvement. This mechanism aggregates and updates pupil models based on a communal dataset and global anchors, ensuring alignment and adaptability to evolving IoT environments. Finally, A group of comprehensive experimentations is conducted to evaluate the performance of FIDWATCH using two demonstrative datasets (BoT-IoT and ToN-IoT). It is noteworthy to mention here that the technique introduced in FIDWATCH can be adapted to enhance security measures in Ad Hoc networks, which share similar decentralized and dynamic characteristics with IoT environments. The results demonstrate the competent detection abilities over a wide range of IoT security threats. The results show that the FIDWATCH can be integrated as a trustworthy and dependable tool for detecting cyber-attacks.

The remainder of the paper is structured as follows and is shown in Fig. 1. In Section 2, the background and literature are surveyed and debated. The system design is presented in Section 3. Then, the methodology of the proposed FIDWATCH is described and explained in Section 4. The experimental details are given in Section 5. Section 6 discusses the experimental results obtained from proof-of-concept simulations. Opportunities and challenges are presented in Section 7, with conclusions in

Section 8.

2. Background & literature

In light of the rapid proliferation of IoT devices over the past few years, academics and researchers have been devoting a significant amount of effort toward improving the ability of existing systems to detect a broad range of IoT security threats. Ioannou et al. [3] applied a power-efficient mL approach to categorize dissimilar kinds of attacks, such as Man-in-the-Middle (MitM) and Distributed Denial of Service (DDoS), then identify previously unknown attacks using retraining using FL to make the cloud server remain updated with the latest model changes while keeping data privacy and security. Su et al. [4] proposed an Adaptive Personalized Federated learning (APFed) to enable many MMSN owners to be involved in cooperative training through the integration of an adapted tailored update and a common global model, which helped to avoid the opposing properties of imbalanced, Non-Independent, and Identically Distributed (Non-IID) data, within IDS. Besides, Rabieinejad et al. [5] introduced a new two-level privacy-preserving method that integrated FL with partially homomorphic encryption for reliable detection of security threats while keeping a good balance between computing complexity as well as detection performance. The proposed method enabled the centralized server to perform encrypted aggregation of local updates without decryption, thereby conserving complex information from possible exposures. Additionally, Doriguzzi-Corin et al. [6] developed adaptive Federated Learning (FLAD) for detecting DDoS attacks with the help of an adaptive method

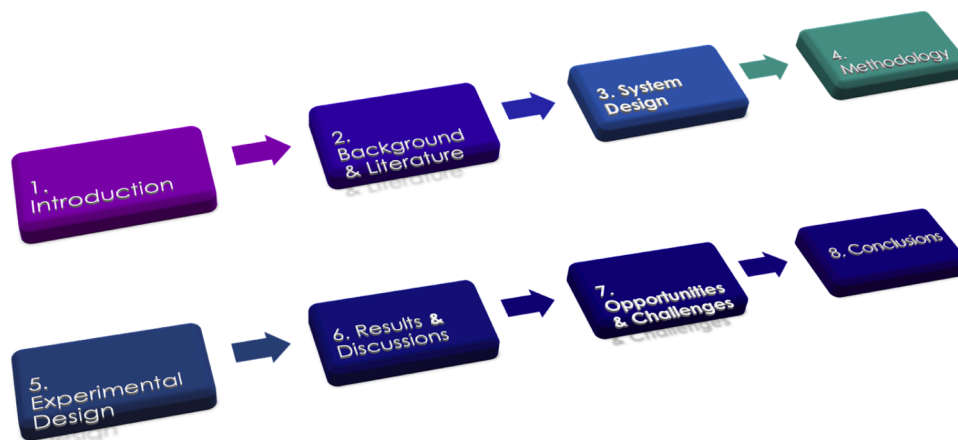


Fig. 1. Visualization of the structure of the paper in different sections

for orchestrating the learning procedure by energetically transferring more calculations to the clients with hard-to-learn attack profiles without distributing test info to screen the performance of the trained model.

Bukhari et al. [7] developed Stacked Convolutional Neural Network and Bidirectional Long Short Term Memory (SCNN-Bi-LSTM) to detect intrusions in a wireless sensor network. The authors emphasized that the SCNN-Bi-LSTM model is trained to efficiently categorize sophisticated and formerly unknown cyber threats by accurately inspecting both local and temporal connections in traffic flows. The model has been specifically designed to detect and categorize different types of Denial of Service (DoS) attacks using specialized WSN-DS and CIC-IDS-2017 datasets. Compared to traditional Artificial Deep Neural Network (ADNN) models, our proposed FL-SCNN-Bi-LSTM model demonstrated superior detection rates for complex and unknown attacks, significantly improving IDS performance. Jin et al. [8] presented a novel IDS framework, named FL-IIDS to successfully discourse the disastrous disremembering problem. The authors proposed a custom loss function for updating the parameters during local training by assigning the learning weights to data of different classes, making the learning proportion of the new classes decrease and deepening the memory of the remaining classes. Salim et al. [9] developed an information fusion-driven CTI framework for securing industrial IoT environments against different cyber-attacks. They design an inclusive cyberattack dataset with efficient feature selection. The authors introduced an ANN-based FL model to decrease the model training rounds according to fulfillment level and estimate the nonappearance of enhancement in the mean accuracy of each attack vector.

The literature also considers blockchain technologies as an essential tool to facilitate the detection and mitigation of IoT security threats either independently or combined with AI. For instance, Basset et al. [10] introduced a blockchain-orchestrated edge intelligence (BoEI) framework, in which a decentralized FL (called Fed-Trust) is applied to learn in a semi-supervised manner to detect cyberattacks in industrial IoT. It included a reputation-based ledger for decentralized recording and authentication of the transactions in a way that maintains the security and privacy of training information [11]. In [12], Sarhan et al. developed a hierarchical blockchain-based FL to build a protected and privacy-preserved collaborative IDS. This highlighted the significance of sharing privacy-preserved threat intelligence among inter-organizational IoT networks, where smart contracts were applied to verify the conformance of executed learning tasks. He et al. [13] developed a conditional generative adversarial net (CGAN)-based collaborative detection of intrusions within blockchain-empowered distributed FL. The long short-term memory (LSTM) is integrated to improve the feature extraction capability, while the blockchain ledger was applied to store and share the training models to guarantee security during the global model's aggregation. More recently, Ali et al. [14] reviewed the literature comprehensively with a main emphasis on the integration of Blockchain and FL to advance and improve the detection of security and privacy threats, mainly those related to industrial IoT networks. Their survey introduced a group of recommendations, a set of challenges, and future research directions that can promote further explore the fusion of these two technologies for developing more trusted IDS for IoT.

Compared to the existing approaches such as [3], which rely on retraining using federated learning to keep the cloud server updated, FIDWATCH reduces the need for frequent training by proposing a contrastive focal loss to enhance discrimination of diverse threats and emphasize hard-to-classify samples. It also uses a fine-grained Knowledge Distillation (KD) to create a compact pupil model, improving efficiency and privacy. Furthermore, FIDWATCH's adaptive incremental updating method allows the global model to refine itself based on shared data, enhancing its ability to handle dynamically evolving security threats in contrast to the methods proposed in [7],[9]. For example, the method in [4] addresses imbalanced and non-IID data but might struggle

with dynamically evolving security threats. In contrast to the methods [5-6], FIDWATCH's KD technique and adaptive updating can reduce computational complexity and are suitable for resource-constrained environments.

3. System design

This section presents a holistic delineation of the system design intended for the proposed FIDWATCH. It includes the system model defining its main architecture and building components, as well as the threat model, which defines the range of possible security vulnerabilities threatening the security and privacy of IoT ecosystems.

3.1. System model

The proposed method uses globally dispersed edge devices and cloudlets to provide cooperative training of the proposed malware detector, as opposed to DL, which is centralized and maintained on either local devices or cloud servers. This is achieved independently of any DL infrastructure, whether on-premises or in the cloud. Fig. 1 shows the architecture of the proposed method for the edge-fog-cloud IoT ecosystem. There are three levels to our system's model. The cloud tier is also referred to as the coordination node in some circles. It includes the cloud servers that are responsible for initializing the hyperparameters, training the initial model, and broadcasting the trained model to the fog nodes. During the training rounds, it compiles the results from the various local classifiers and then adjusts the parameters of the global classifiers to reflect these changes [15].

The fog tier is an intermediate level of computation that sits between the edge tier and the cloud. It is made up of fog nodes, which perform this function. In comparison to edge devices, it possesses significantly more powerful computational resources. These nodes are where the confidential and sensitive data that has been compiled from the relevant smart city sector is stored. Following the receipt of the initial classifier from the cloud tier, each fog device initiates the process of local model training utilizing its confidential data. After that, they aggregate the results of the distributed training by uploading the updated local parameters to a server that is hosted in the cloud. This procedure is carried out once again after each communication round-up until the point where the global model converges.

End users with constrained resources utilize the devices that make up the Edge Tier. The federated training uses data generated by these devices. These devices come in a wide range of varieties and produce various amounts of data with varied levels of confidentiality. These data are not transferred over the IoT network but are instead locally held on the fog node that has been allocated to them. Private data, such as health records and corporate information, may be among them.

3.2. Threat model

The growing adoption of IoT and the great shift incurred in communications technologies have been broadening the threat surfaces to an unprecedented range. In this regard, developing a reliable framework for the detection of IoT security threats makes it essential to interpret potential security threats surrounding IoT systems. Accordingly, the threat model devised for the proposed system (as reviewed in Table 1) provides a comprehensive description of different families of IoT security threats compromising the truthfulness, confidentiality, and accessibility of the IoT systems. As shown in Table 1, the main families of IoT security threats include physical attacks, Network-based threats [16], Software-based threats [17], Data-Related threats [18], Social Engineering threats [19], Supply Chain threats [20], and Cryptojacking threats [21]. As the name implies, physical threats represent a tangible threat to the physical layer of IoT architecture, thereby requiring it to lie in the proximity of IoT devices. This family includes but is not limited to different device tampering and side-channel vulnerabilities. Besides,

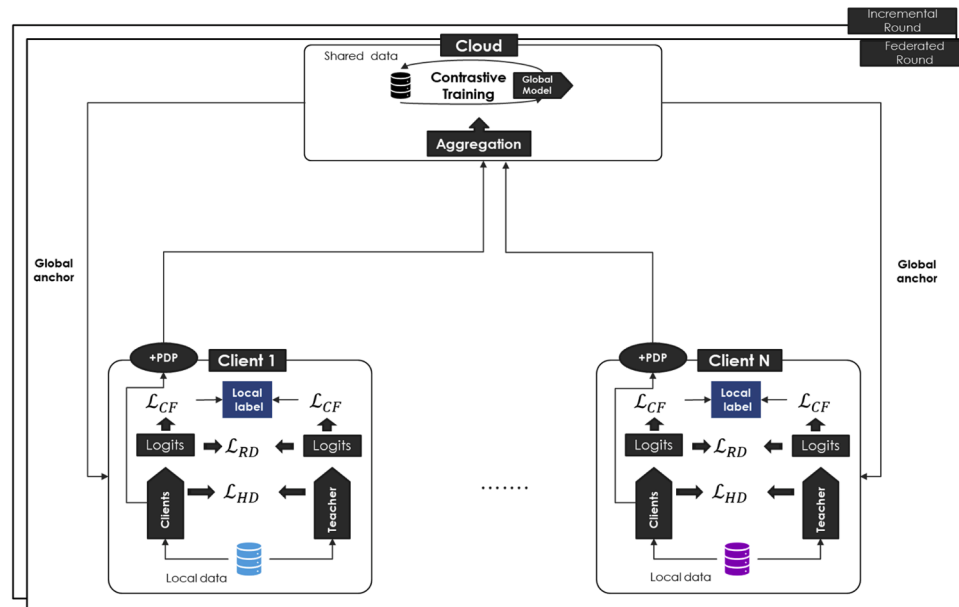
Table 1

A summary of IoT security threats used in our threat model.

Security Threats Family	Attack	Description	Target Layer(s)	Severity	Frequency	Legal Implications
Physical Threats	Tampering	Physically manipulate IoT devices to compromise integrity	Edge, Device	High	Low: Mid	Property damage, privacy violations
	Side-Channel Attacks	Exploit secrets from IoT devices through analysis of physical parameters (e.g., Cache, Timing, Power analysis, Electromagnetic).	Edge, Device	Medium to High	Moderate	Data theft, privacy violations
Network-Based Attacks	Denial of Service (DoS)	Flood network or device to disrupt operations	Network	High	High	Service disruption, financial losses
	Man-in-the-Middle (MitM)	Intercept and alter communication between devices	Network	High	Moderate	Data interception, privacy violations
	Packet Sniffing	Unauthorized capture and analysis of data packets	Network	Medium to High	Moderate to Low	Data theft, privacy violations
Software-Based Attacks	Malware	Software designed to infiltrate, damage, or gain unauthorized access	Edge, Device	High	High	Data theft, financial losses
	Remote Code Execution (RCE)	Exploit vulnerabilities to execute arbitrary code remotely	Edge, Fog, Cloud	High	Moderate to High	Unauthorized system access, data theft
	Zero-Day Exploits	Leveraging unknown vulnerabilities for malicious purposes	Edge, Fog, Cloud	High	Low	Unauthorized system access, data theft
	Firmware Attacks	Targeting vulnerabilities in device firmware	Edge, Device	High	Moderate to Low	Unauthorized system access, data theft
Data-Related Attacks	Data Breaches	Unauthorized access or disclosure of sensitive data	Network, Cloud	High	High	Data theft, regulatory fines
	Data Manipulation	Unauthorized alteration of transmitted or stored data	Network, Cloud	High	High	Data corruption, financial losses
Social Engineering Attacks	Phishing	Deceptive techniques to trick users into revealing sensitive information	Network, Cloud	Medium to High	High	Unauthorized data access, financial losses
	Impersonation	Pretending to be a legitimate entity to gain trust and access	Network, Cloud	High	High	Unauthorized data access, financial losses
Supply Chain Attacks	Counterfeit Components	Inclusion of counterfeit components during manufacturing	Device	High	Moderate	Device malfunction, security vulnerabilities
	Third-Party Compromise	Compromising third-party services involved in the supply chain	Network, Cloud	High	High	Unauthorized data access, financial losses

network-based threats seek to compromise device connectivity or intercept the transmitted sensitive data by exploiting the vulnerabilities in network communication protocols and channels. As the name implies, software-based security threats focus on significant threat vectors that exploit the vulnerabilities in device firmware or software applications. These threats enable hackers to gain unauthorized access, execute arbitrary code, or engage in other malicious activities to achieve objectives against the IoT system. For example, firmware attacks are a popular form of attack to compromise the firmware of IoT devices,

aiming to interrupt their normal functionality. When it comes to data, security threats turn out to be more critical and have catastrophic consequences, as data-related attacks always seek to expose sensitive information within IoT systems for cybercrime and thefts. Moreover, to obtain unauthorized entry or to obtain sensitive information, social engineering tactics take advantage of human weaknesses to violate the security of systems and data. Supply chain threats represent the ability to exploit the vulnerabilities of services and IoT devices that occur in the course of the manufacturing or integration process, which might lead to

**Fig. 2.** Visualization of general workflow of the proposed FIDWATCH

cascading consequences in the complete infrastructure. Table 1 shows each attack type used within our threat model, along with targeted IoT layers. Also, we provide useful insights regarding the severity level of each attack, the frequency rate, and corresponding implications and consequences.

4. Methodology of FIDWATCH

This section presents a detailed discussion of the methodology of the IDS presented in this paper. We begin by introducing a formulation of the problem used in this work. Then, methodological details of the building blocks of the FIDWATCH framework are presented, which provides valuable theoretical insights regarding the implications of incremental distillation on the ability to monitor and identify IoT security threats (see Fig. 2).

In a complex IoT ecosystem, the security traffic flows are characteristically localized to distinct clients, and sharing these traffic flow data with other clients is severely barred. The primary aim of FIDWATCH is to learn from an endlessly developing stream of data that presents new types of security threats in extremely distributed environments. The traffic flows are alienated into an arrangement of non-overlapping training tasks, each on behalf of a phase of incremental learning. The ultimate goal is to continually develop an intrusion detection model that distills knowledge from both existing and preceding knowledge across countless training nodes. After each knowledge acquisition, the model's performance is assessed on all types of security threats it has encountered. These include the types of security threats it has learned from earlier representational knowledge. We are presenting the design of FIDWATCH to solve the above-mentioned problem. It is based on a multi-level iterative process for incremental updating of the global detection model. This updating is done based on the knowledge distilled from locally trained models hosted by different clients in an IoT environment. This, in turn, empowered the FIDWATCH to gradually refine its detection capabilities over time. Each incremental round concentrates on particular updates and adaptations according to the evolving variation in the distribution of IoT datasets and underlying security threats.

The initial parameters of the global detection model m with parameters Θ_g are determined during the initialization phase of FIDWATCH. This includes a complete definition of model architecture in terms of network layers, activation functions, and dropout rate. The model architecture is carefully designed to accommodate the complexities of IoT security threat detection tasks while ensuring scalability and compatibility across diverse IoT environments. This includes determining hyperparameters like learning rate ($\eta = 0.001$), batch size ($B = 32$), and regularization terms (λ) governing the optimization of parameters during the training process. In addition, multiple elementary specifications are initialized to begin the incremental training process in terms of the number of communication rounds (R_{comm}) and incremental rounds (R_{inc}), deciding the frequency and granularity of local updates and aggregation. Each incremental round consists of multiple federated rounds, with the number of selected clients (K) dynamically adjusted based on availability and reliability. Each client k contribute their local datasets to the federated training using local data D_k composed of m_k data points sampled from the distribution P_k .

By the start of each incremental round in FIDWATCH, the clients are selected based on immediate feedback and system dynamics, guaranteeing the adaptive and effective contribution of the clients to the overall learning process. In other words, the training coordinators evaluate the contemporary system state and dynamically regulate the underlying selection criteria to optimize the detection capability of FIDWATCH. This set of criteria includes a variety of factors such as data distribution and quality, network availability, resources vector, and historical performance. These factors are mathematically symbolized as D_k, Q_k, N_k, R_k , and H_k , respectively, for each client k in the client pool. Then, FIDWATCH dynamically selects clients in a way that maximizes both system

fairness criteria and detection performance. This optimization process can be expressed as:

$$\mathcal{K} = \underset{\mathcal{K}}{\operatorname{argmax}} f(D_k, Q_k, N_k, R_k, H_k), \quad (1)$$

wherein $f(\dots)$ represent the objective function representing a weighted combination of partaking of various selection criteria.

$$f(D_k, Q_k, N_k, R_k, H_k) = \sum_{k=1}^K w_k \cdot g(D_k, Q_k, N_k, R_k, H_k) \quad (2)$$

In the above terms, the w_k denote the weight assigned to k_{th} client, while $g(\dots)$ symbolize the function mapping the selection criteria into a scalar value of contributions. With this real-time feedback about clients, FIDWATCH can regularly adapt its behaviour to cope with dynamic system conditions and maintain a robust ability to detect IoT security threats.

The selected local clients experience supervised learning from the local IoT security data, D_k , where the client's data are kept consistent during all federated rounds. In each round, the local models update their parameters by optimizing a contrastive focal loss, $\mathcal{L}_{CF}$, which is proposed to concurrently improve the feature discrimination and detection capabilities. In mathematical terms, the local data contain pairs of data elements $\{(x_i, y_i)\}_{i=1}^{m_k}$, in which x_i and y_i denote the input IoT flow, and label, respectively. The local learning used the data D_k to optimizes the $\mathcal{L}_{CF}$ objective function, and is given below:

$$\mathcal{L}_{CF}(D_k) = \frac{1}{m_k} \sum_{i=1}^{m_k} [(1 - y_i) \cdot \mathcal{L}_{focal}(p_{i,0}, y_i) + y_i \cdot \mathcal{L}_{contrastive}(f(x_i), y_i)], \quad (3)$$

In the above formula, the $\mathcal{L}_{focal}(p_{i,0}, y_i)$ denote the categorical focal loss for the negative class (class 0) containing normal or benign samples.

$$\mathcal{L}_{focal}(p_{i,0}, y_i) = -(1 - p_{i,0})^\gamma \cdot y_i \log(p_{i,0}), \quad (4)$$

Wherein $p_{i,0}$ denote the predicted probability that i_{th} sample belongs to the negative class. γ is the tunable concentration parameter that controls the prominence of hard-to-classify instances. Besides, $f(x_i)$ symbolize the feature vector attained via the local detection model for i_{th} sample. On the other hand, $\mathcal{L}_{contrastive}(f(x_i), y_i)$ represent the contrastive loss for the positive class

(class 1), containing IoT attacks, and it is represented as follows:

$$\mathcal{L}_{contrastive}(f(x_i), y_i) = \frac{1}{2} (1 - y_i) \|f(x_i) - \mu_1\|^2, \quad (5)$$

With μ_1 representing the centeriod of class 1. Throughout the local learning iterations, each selected client k iteratively upgrade the parameters of the local detection model Θ_k by optimizing the above $\mathcal{L}_{CF}$ using AdamW optimizer shown in Algorithm 1. Through conjoint optimization of the model parameters based on $\mathcal{L}_{CF}$, the local classifier becomes capable of capturing discriminatory features and advancing the detection abilities for IoT attacks. This supports the ultimate goal of improving effectiveness in threat detection.

Upon the completion of local learning in FIDWATCH, Knowledge distillation [23-25] come into action to transfer knowledge from the local model, likely to be composite and computationally intensive, to a lighter, more compact architecture, called a pupil model. This in turn can assist process deploying efficient detection models on resource-constrained IoT devices while preserving high performance.

A common limitation of the existing federated distillation approach lies in transmitting a small amount of information from the clients to the data because the logits of the model are only info communicated with the global coordinator. In an attempt to address this downside, a fine-grained KD is designed in FIDWATCH, in which each client k communicates some intermediary representations instead of the model's output. In other words, a pupil model is trained to mimic the conduct of the local detection model, capturing its learned representations and decision-making processes. The distillation operation is guided by a

Algorithm 1

Optimization of local params with AdamW optimizer [22]

```

input:  $\gamma(\mathbf{l}_r), \beta_1, \beta_2, \theta_0, f(\theta)$  (objective),  $\epsilon, \lambda, \text{agd}, \max$ 
1 initialize:  $m_0 \leftarrow 0$  (first moment),  $v_0 \leftarrow 0$  (second moment),  $\widehat{v}_0^{\max} \leftarrow 0$ 
2 for  $t = 1$  to ... do
3   if  $\max: g_t \leftarrow -\nabla_{\theta} f(\theta_{t-1})$ 
4   else
5      $g_t \leftarrow \nabla_{\theta} f(\theta_{t-1})$ 
6      $\theta_t \leftarrow \theta_{t-1} - \gamma \lambda \theta_{t-1}$ 
7      $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$ 
8      $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$ 
9      $\widehat{m}_t \leftarrow \frac{m_t}{(1 - \beta_1^t)}$ 
10     $\widehat{v}_t \leftarrow \frac{v_t}{(1 - \beta_2^t)}$ 
11    if agd
12       $\widehat{v}_t^{\max} \leftarrow \max(\widehat{v}_t^{\max}, \widehat{v}_t)$ 
13       $\theta_t \leftarrow \theta_t - \frac{\gamma \widehat{m}_t}{(\sqrt{\widehat{v}_t^{\max}} + \epsilon)}$ 
14    else
15       $\theta_t \leftarrow \theta_t - \frac{\gamma \widehat{m}_t}{(\sqrt{\widehat{v}_t} + \epsilon)}$ 
16 return  $\theta_t$ 

```

distillation loss function, which promotes the pupil model to achieve a set of outputs that is almost similar to those obtained by the teacher while achieving a smaller model size.

In FIDWATCH, the proposed fine-grained KD is built using a careful combination of multiple losses for optimizing the teacher and pupil models locally, which include $\mathcal{L}_{CF}$ loss for local supervision, a reciprocated distillation loss to communicate knowledge from output soft labels, and an adaptive hidden loss to purify knowledge from the hidden states. The joint loss function for KD is given by:

$$\mathcal{L}_{KD} = (1 - \lambda_1 - \lambda_2) \cdot \mathcal{L}_{CF}(D_k, \Theta_p) + \lambda_1 \cdot \mathcal{L}_{MD}(D_k, \Theta_t, \Theta_p) + \lambda_2 \cdot \mathcal{L}_{HD}(D_k, \Theta_t, \Theta_p), \quad (6)$$

where $\mathcal{L}_{RD}(D_k, \Theta_t, \Theta_p)$ estimate the divergence between the softened probabilities generated by the teacher model Θ_t and the pupil model Θ_p using the same set of input. $\mathcal{L}_{HD}(D_k, \Theta_t, \Theta_p)$ measure the loss between hidden states of the teacher model Θ_t and the pupil model Θ_p . λ_1 and λ_2 are hyperparameters that regulate the trade-off between the different loss terms. The adaptive mutual distillation loss $\mathcal{L}_{MD}$ is definite as the Kullback-Leibler (KL) divergence between the softened probabilities generated from both teacher and pupil denoted as $P_{\text{soft}}(\Theta_t)$ and $P_{\text{soft}}(\Theta_p)$, respectively.

$$\mathcal{L}_{MD}(D_k, \Theta_t, \Theta_p) = \text{KL}(P_{\text{soft}}(\Theta_t), P_{\text{soft}}(\Theta_p)) \quad (7)$$

The $\mathcal{L}_{HD}$ in (7) is defined as the mean squared error (MSE) between the hidden states of the teacher model and the pupil model:

$$\mathcal{L}_{HD}(D_k, \Theta_t, \Theta_p) = \frac{1}{N} \sum_{i=1}^N \|h_i^t - h_i^p\|^2 \quad (8)$$

Note that the h_i^t and h_i^p in (8) denotes the hidden states of the teacher and pupil, respectively, while N is the number of hidden states.

A probabilistic DP is proposed in FIDWATCH for protecting the privacy of the local data against possible leakage while maintaining clients as well as quantifying and limiting the privacy loss (see Fig. 2). To this end, differential privacy refers to a randomization mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ whose output on somewhat input database d is not distinct from the output on the somewhat adjoining database d' varying from d in single data points [26–28]. For any part of the outputs $S \subset \mathcal{R}$, this can be expressed as:

$$P[\mathcal{M}(d) \in S] \leq \exp(\epsilon) P[\mathcal{M}(d') \in S] + \delta, \quad (9)$$

The DP of a method $\mathcal{M}$ is attained through restraining the relative sensitivity as follows,

$$\Delta(\mathcal{M}) = \max_{d_1, d_2 \in \mathcal{D}} \|\mathcal{M}(d_1) - \mathcal{M}(d_2)\| \quad (10)$$

and then put on a randomized noise machinery. With the use of the pre-trained feature extractor h_0 per each client, a logistic regressor is applied to discriminate between the local attacking data D_i and the negative data D^- through optimization of regularized empirical risk minimization (ERM) objective [29]:

$$w_i^* = \arg\min_w J(w, h_0, D_i, D^-) \quad (11)$$

where,

$$J(w, h_0, D_i, D^-) = a \sum_{x \in D_i \cup D^-} l(t_x(w, \widetilde{h}_0(x)) + \lambda R(w) \quad (12)$$

$$t_x = 2(1_{x \in D_i}) - 1 \in [-1, 1], a = -, \widetilde{h}_0(x) = h_0(x) (\max_{x \in D_i \cup D^-} \|h_0(x)\|)^{-1} \quad (13)$$

whereas t_x expresses the ground truth for the classification task, a and $\widetilde{h}_0(x)$ denote the normalization factor as well as normalized features, correspondingly. $R(w)$ is ℓ_2 -regularizer and is defined as $R(w) = \left(\frac{1}{2}\right) \|w\|_2^2$. Solving the regularized ERM objective w_i^* , the inevitability scores on the distillation data D_{distill} is attained through the logistic probability, defined as:

$$p_i(x) = \frac{1}{1 + \exp(-\langle w_i^*, \widetilde{h}_0(x) \rangle)} + \xi. \quad (14)$$

In (14), $\xi = 5e - 9$. denote constant to guarantee numerical stability. Given that l is differentiable with $|l'(z)| \leq 1 \forall z$, and $R(\cdot)$ is convex and differentiable, then the ℓ^2 sensitivity $\Delta_2(\mathcal{M})$ of the mechanism, given by:

$$\mathcal{M} : D_i \mapsto \arg\min_w J(f, h_0, D_i, D^-) \leq 2(\lambda(|D_i| + |D^-|))^{-1} \quad (15)$$

Indeed, the sensitivity scales contrariwise with the magnitude of the entire data $|D_i| + |D^-|$. With the injection of the Gaussian noise in (15), the sanitary edition of the randomized technique can be derived as follows:

$$\mathcal{M}_s : D_i \mapsto \arg\min_f J(f, h_0, D_i, D^-) + N \quad (16)$$

In simulations, the privacy analysis emphasizes only the ranking

function. The DP-based training of local models f_i is beyond the scope of this paper, and presented in literature studies. In [30], the composite privacy cost of applying FIDWATCH is computed as a combination of the privacy loss from both ranking function $\mathcal{M}_s$ as well as communicating the local update updated f_i , which is the common case for all FL settings.

In FIDWATCH's incremental approach, a joint dataset is used as a reference point to address data heterogeneity among clients, following common practice in federated learning literature. This joint dataset is accessible to all clients and serves as a standardized set of samples for model evaluation and alignment. At the end of each federated round, every client generates model outputs for each sample in the joint dataset. These outputs are transmitted to the central server and averaged across all clients to obtain global model outputs. The aggregation process for the pupil models is then performed using the global model outputs as references, ensuring alignment between the clients' models and the global model. The aggregated pupil model $\Theta_{p,agg}$ at the central server is computed as follows:

$$\Theta_{p,agg} = \sum_{k=1}^N w_k \cdot \Theta_{p,k} \quad (17)$$

where N refers to a number of clients. w_k refers to the weight assigned to the client k based on its performance on the shared dataset. The weight w_k for each client, k is determined based on its model performance on the joint dataset, which serves as a proxy for data heterogeneity. Clients that perform well on the joint dataset contribute more to the aggregated model, reflecting their ability to generalize to unseen data. Additionally, the global model outputs from the shared dataset serve as global anchors for knowledge distillation in the subsequent federated round. Each client performs knowledge distillation using these global anchors, ensuring that the knowledge learned from the joint dataset is aligned across all clients and with the global model.

5. Experimental design

In this section, details of the simulations that were conducted are presented to provide useful insights about the design of our experiments and a holistic guide that facilitates the reproducibility of our work.

5.1. Experimental setup

The experimentations of this work are performed on a workstation device with Intel (R) Xeon (R) CPU E5-2670 0@ 2.60GHz processor, 128 GB memory, NVIDIA Quadro Accelerator. A Windows 10 64-bit system has been adopted as an operating system for our device. Pytorch 1.8 libraries were adopted to implement and develop all models within a Python 3.9.0 environment. In our experiments, an FL is considered with a total of $n = 100$ joining clients, in which the client contribution rate varies from 20% to 100%. Dirichlet distribution [30] is used for dividing the IoT training data regularly among the clients, which promotes smooth adaption of the degree of non-i.i.d.-ness in the client data using the Dirichlet variable α , whose values vary from 100.0 and 0.01. Equal label distribution is determined with $\alpha = 100$, while $\alpha = 0.01$ indicate that the majority of data on every client belongs to one type of IoT attack.

5.2. Performance metrics

This paper uses common metrics to evaluate the detection performance of the proposed approach. The common metrics are based on the quartiles, that is false negative (FN), true negative (TN), false positive (FP), and true positive (TP). These metrics are mathematically expressed as follows:

$$Precision = \frac{TP}{TP + FP} \times 100 \quad (18)$$

$$Recall = \frac{TP}{TP + FN} \times 100 \quad (19)$$

$$F1 - score = 2 * \frac{Recall * Precision}{Recall + Precision} \quad (20)$$

5.3. Dataset description

In the experimental validation, the efficacy of our proposed FIDWATCH framework is evaluated using two different but complementary datasets reflecting exclusive outlooks about the dynamics of cybersecurity threats in IoT systems. In this section, we provide a comprehensive overview of these datasets, detailing their sources, characteristics, and methodologies employed for data collection and analysis. The first dataset with 35,000,000 samples is designated by generating the NetFlow records from the public pcaps files of the BoT-IoT dataset. The traffic flows are annotated using the corresponding types of attacks. Among them, 99.4% of data are attacking samples, and the remaining is normal traffic samples. The class distribution of this dataset is shown in Fig. 3.

The second dataset with 17,000,000 samples is created by generating the NetFlow records from the public pcaps files of the ToN-IoT dataset. Among them, 60% of data are attacking samples, and the remaining are normal traffic samples. The class distribution of this dataset is shown in Fig. 4. Raw data undergoes a few preparation procedures to ensure effective model training. First, the label-encoding approach is used to encode the values of the categorical variables into numerical information. Additionally, as shown in (21), min-max normalization is used to bring the feature values into the range of [0,1] to mitigate the impact of large disparities between the values of various features.

$$x_{normalized} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (21)$$

The symbols x , x_{max} , x_{min} signify the current feature value, the minimum feature value, and the lowest feature value. Adaptive synthetic sampling (ADASYN) technique [31] is applied to generate artificial samples for minority classes in the dataset.

6. Results and discussions

In this section, we systematically discuss the empirical results obtained from the experiments that were conducted, along with the related discussions, analyses, and implications.

6.1. Comparative analysis

The performance of FIDWATCH is evaluated and compared against cutting-edge models, which include LSTM [14], CNN-GRU [15], FED-IDS [9], FML [16], Fed-Trust [8], FL-CTIF [7], FL-IIDS [6], and FLAD [4], using a different set of evaluation measures (mentioned in Section 5). The comparative results are measured and reported in terms of mean $\pm$ standard deviation driven from five self-determining runs, which is a favored approach to get insights about the general performance of the model. The results of all studies are reproduced under the same experimental settings to ensure fair and reliable comparison between the detection performance of models. In addition, the Friedman test ensures the statistical importance of results instead of being achieved by luck.

The comparative results for BoT-IoT data across different evaluation metrics are introduced in Table 2. Compared to centralized models (LSTM [32] and CNN-GRU [33]), FIDWATCH achieved notable performance improvements across all evaluation metrics. This demonstrates the ability of FIDWATCH to maintain the ability to detect IoT security threats with performance overcoming the centralized settings. The performance improvements are further validated with statistical analysis, resulting in p-values of

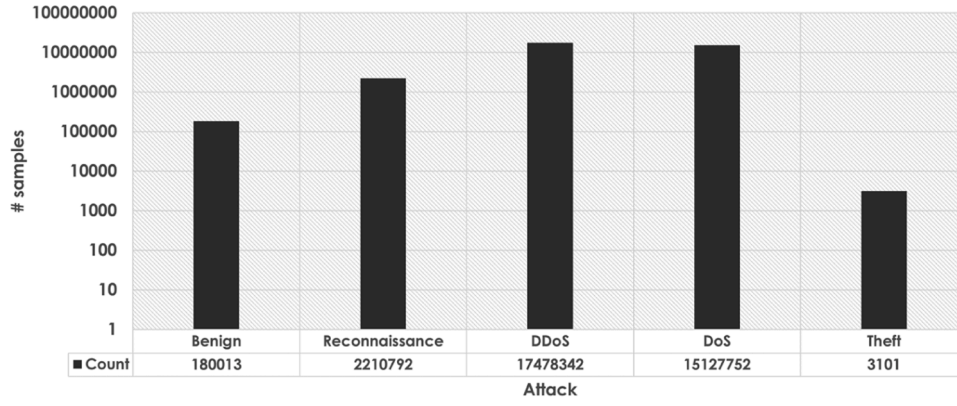


Fig.3. Distribution of samples across classes in BoT-IoT data.

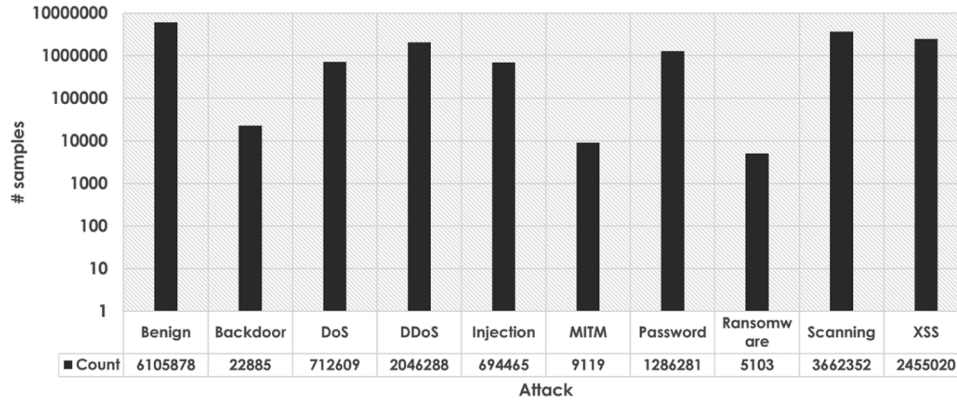


Fig.4. Distribution of samples across classes in ToN-IoT data.

Table 2

Quantitative results obtained from comparing the proposed approach against cutting-edge solutions on BoT-IoT dataset (Mean $\pm$ STD).

Methods	AUC	Precision	F1-score
LSTM [32]	96.37 $\pm$ 1.39	91.24 $\pm$ 2.06	93.8 $\pm$ 4.43
CNN-GRU [33]	97.04 $\pm$ 2.08	90.85 $\pm$ 4.98	91.67 $\pm$ 1.67
FED-IDS [11]	96.33 $\pm$ 2.04	89.88 $\pm$ 3.99	89.47 $\pm$ 3.95
FML [34]	97.27 $\pm$ 1.15	89.71 $\pm$ 3.78	92.62 $\pm$ 0.01
Fed-Trust [10]	97.54 $\pm$ 1.05	<u>96.1$\pm$2.32</u>	94.89 $\pm$ 2.65
FL-CTIF [9]	98.82 $\pm$ 1.14	92.63 $\pm$ 1.53	94.19 $\pm$ 0.24
FL-IIDS [8]	<u>99.07$\pm$1.08</u>	95.56 $\pm$ 2.86	<u>95.04$\pm$0.69</u>
FLAD [6]	97.28 $\pm$ 2.03	90.32 $\pm$ 3.76	93.81 $\pm$ 1.57
FIDWATCH	99.64$\pm$0.81	97.27$\pm$1.5	97.07$\pm$0.66

6.364E-06, and 4.506E-08, respectively. When it comes to comparisons with FED-IDS [11], and FML [34], FIDWATCH overcomes their detection performance with significant margins. This validates the contribution of proposed incremental distillation to enhancing the ability of FIDWATCH to detect and mitigate IoT security threats. The abovementioned models' observed improvements are further corroborated using statistical testing, yielding p-values of 5.518E-07 and 8.077E-05, respectively. Likewise, the detection performance of FIDWATCH overcame the performance FL-CTIF [7], FL-IIDS [6], and FLAD [4] with mean improvements in the range of 1% to 6% in precision, 2% to 4% in F1-score and 1% to 2% in AUC, which reflect the forcefulness and efficiency of FIDWATCH compared to cutting-edge FL-based intrusion detection approaches. Again, the statistical test obtained a p-value of 7.110E-07, 2.792E-06, and 1.146E-08, respectively, which further confirmed the validity of the improvements achieved.

Table 3 presents the comparative findings for ToN-IoT data across

Table 3

Quantitative results obtained from comparing the proposed approach against cutting-edge solutions on ToN-IoT dataset (Mean $\pm$ STD).

Methods	AUC	Precision	F1-score
LSTM [32]	98.09 $\pm$ 1.19	94.4 $\pm$ 3.15	94.01 $\pm$ 1.28
CNN-GRU [33]	97.55 $\pm$ 2.96	93.92 $\pm$ 1.41	<u>94.98$\pm$4.92</u>
FED-IDS [11]	<u>98.79$\pm$2.63</u>	91.64 $\pm$ 1.21	93.91 $\pm$ 0.68
FML [34]	96.69 $\pm$ 1.71	92.32 $\pm$ 1.8	92.31 $\pm$ 3.13
Fed-Trust [10]	97.64 $\pm$ 1.34	90.64 $\pm$ 3.25	93.87 $\pm$ 1.19
FL-CTIF [9]	96.95 $\pm$ 2.05	93.51 $\pm$ 2.66	92.11 $\pm$ 3.33
FL-IIDS [8]	97.19 $\pm$ 1.79	94.03 $\pm$ 1.44	94.42 $\pm$ 2.94
FLAD [6]	97.86 $\pm$ 0.95	92.34 $\pm$ 2.69	93.15 $\pm$ 1.71
FIDWATCH	99.02$\pm$0.39	94.93$\pm$0.72	95.63$\pm$3.53

various evaluation metrics. FIDWATCH demonstrated substantial improvements in performance over centralized models (LSTM [14] and CNN-GRU [15]) in all metrics used for evaluation. These results highlight FIDWATCH's ability to continue detecting IoT security threats while outperforming centralized settings. The observed improvements over both models are further corroborated using statistical analysis, yielding corresponding p-values of 2.187E-07 and 1.948E-06. In significant differences, FIDWATCH outperforms FED-IDS [9] and FML [16] in terms of detection performance. Thus, the contribution of the suggested incremental distillation to improving FIDWATCH's capacity to identify and lessen IoT security risks is validated. Statistical testing confirms the reported improvements of the previously indicated models, producing p-values of 1.579E-08 and 1.789E-07, respectively. Similarly, FIDWATCH's detection performance outperformed FL-CTIF [7], FL-IIDS [6], and FLAD [4], with mean improvements ranging from 3% to 6% in precision, 2% to 4% in F1-score, and 1% to 3% in AUC. These results

demonstrate FIDWATCH's strength and effectiveness compared to state-of-the-art FL-based intrusion detection techniques. Although our method exhibits some variability (± 3.53), it can achieve a high mean F1 score of 95.63, reflecting a robust detection performance. This variability can be attributed to the differences in data subsets or stochastic conditions in the training process, which is a popular case in federated settings. Yet again, the statistical test obtains a p-value of $6.598E-07$, $5.250E-06$, and $3.948E-08$, respectively, which further confirms the validity of the achieved enhancements. The corresponding confusion matrices are shown in Fig. 5 to interpret the details of the detection performance of FIDWATCH along with the other models (FL-CTIF, FL-IIDS). The results conform to the tabulated results, where the proposed FIDWATCH maintains the highest precision in detecting the different classes of attacks.

6.2. Convergence analysis

The convergence of FIDWATCH analysis provides valuable insights into the characteristics of the learning process in terms of a relation between detection performance estimated with training loss and the number of communication rounds. Fig. 6 shows the convergence results for the BoT-IoT dataset. The learning curve shows a swift reduction in the loss with an increase in the number of communication rounds.

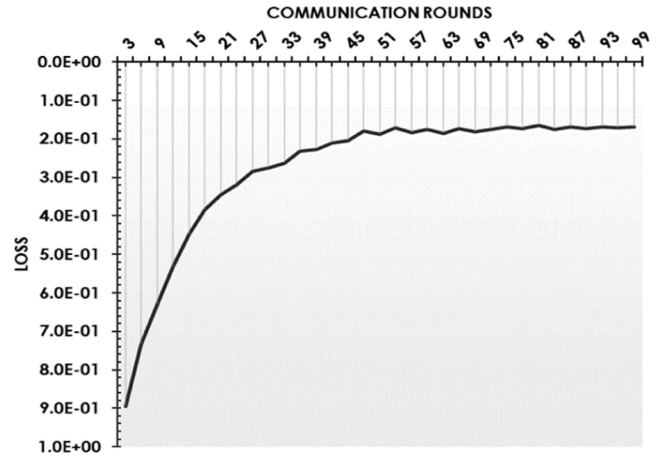


Fig. 6. Convergence analysis for FIDWATCH on BoT-IoT data

Notably, loss reaches a plateau after a certain number of communication rounds, reflecting steady performance after this point. The observed behavior underlines the significance of optimizing the number of rounds to reach the highest detection performance without minimizing

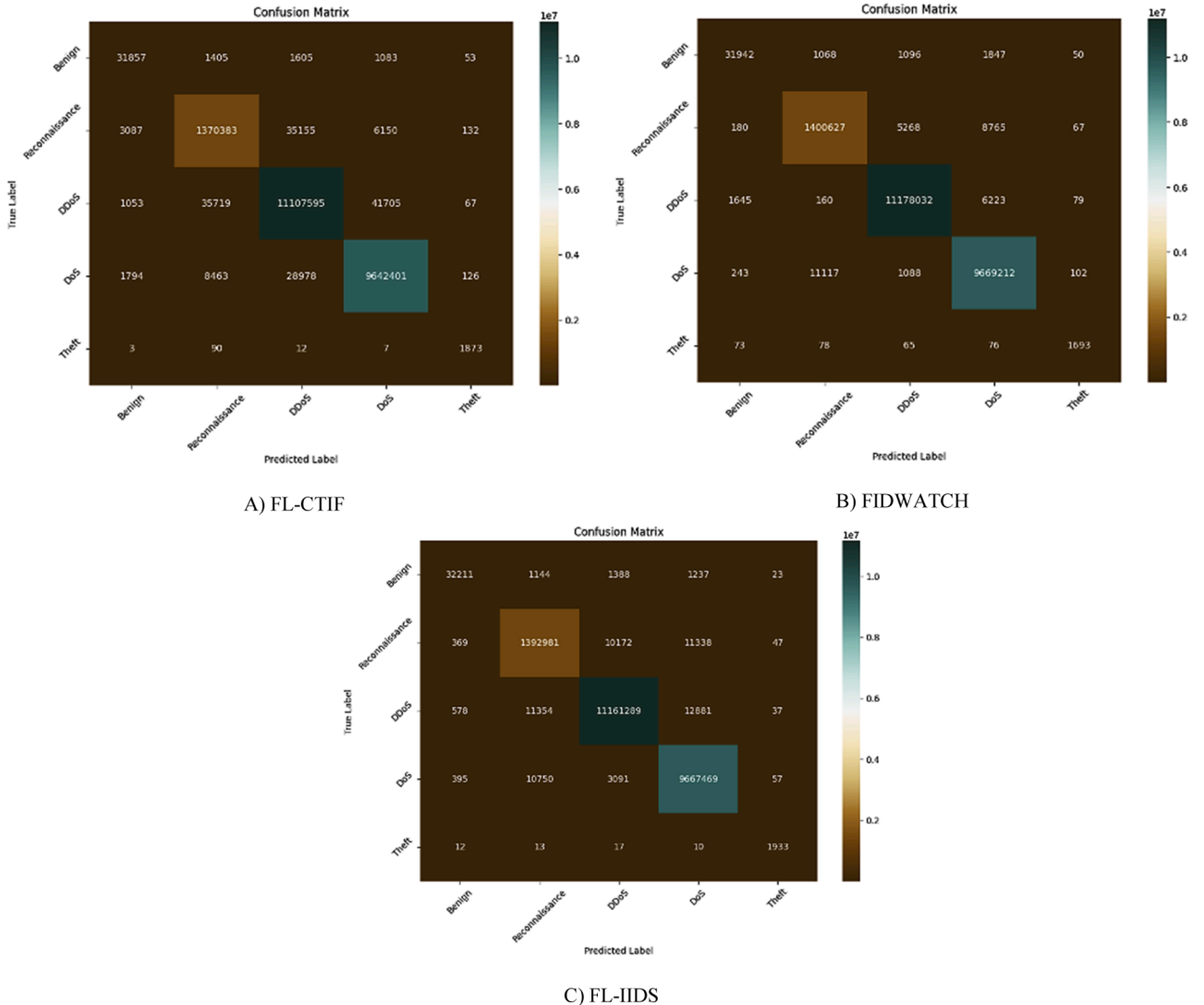


Fig. 5. Comparison between the confusion matrices for top performing approaches on test set of BoT-IoT data.

network overhead. Fig. 7 shows the convergence curve for the FIDWATCH on ToN-IoT dataset. It is worth noting that the learning loss decreases progressively with an increase in a number of rounds, but converges after 30 rounds. Note that the convergence designs might vary because of changes in the attributes of the data in terms of the severity of attacks, data distribution, and network scenarios.

6.3. Scalability analysis

Scalability analysis is conducted to assess the detection abilities of FIDWATCH as the number of participating clients scales up. Fig. 8 and Fig. 9 present the scalability analysis for FIDWATCH on the BoT-IoT and ToN-IoT datasets, respectively. For the BoT-IoT data, the increase in the number of clients did not negatively affect the value of the detection. The F1-score shows that the FIDWATCH can be used in a distributed network of IoT devices to boost threat detection capabilities within the IoT ecosystem. Nevertheless, it's notable that a slight decline in the F1 score after a particular threshold indicates some possible scalability challenges, which might be attributed to data heterogeneity, communication overhead, or limited resources. Fig. 9 shows the scalability of FIDWATCH on ToN-IoT data. It can be observed that the magnitude of the F1 score is kept steady despite increase in a number of clients, which further reflects the ability to scale up FIDWATCH to varied ranges of IoT environments.

6.4. Privacy analysis

This section delves into a detailed analysis of the privacy analysis for FIDWATCH by examining the relation between the detection F1-score and the regularization term as well as privacy parameters given in the previous section. Notably, moderate privacy settings allow for results close to non-private training. Fig. 10 and Fig. 11 present the privacy analysis for FIDWATCH using the BoT-IoT and ToN-IoT datasets, respectively. It is worth noting that there was a significant decrease in F1-score from 97.07% to 84.14% in BoT_IoT data. In ToN_IoT data, the F1-score decreases from 95.63% to 79.89%. Increasing the value of λ can guarantee extra privacy preservation for FIDWATCH, but at a cost of F1-score. These empirical findings explain the rationale behind setting default values of privacy parameters with $\epsilon = 1E - 1$, $\delta = 1E - 5$, $\lambda = 1E - 1$.

7. Open challenges

In the context of security threat detection, a "one model fits all" approach that can learn all image features and identify all vulnerabilities is not currently feasible. Even though there are various well-established

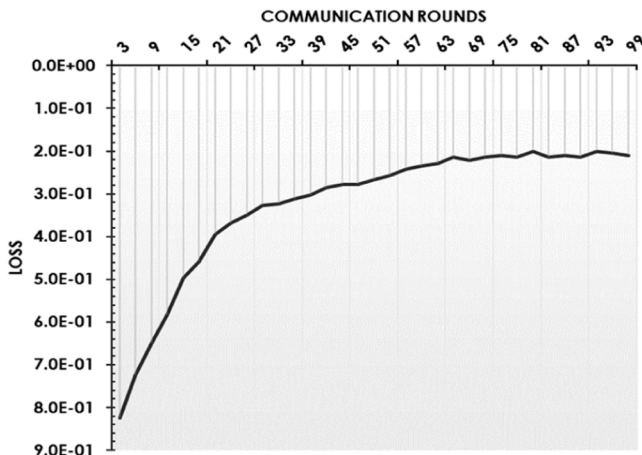


Fig. 7. Convergence analysis for FIDWATCH on ToN-IoT data

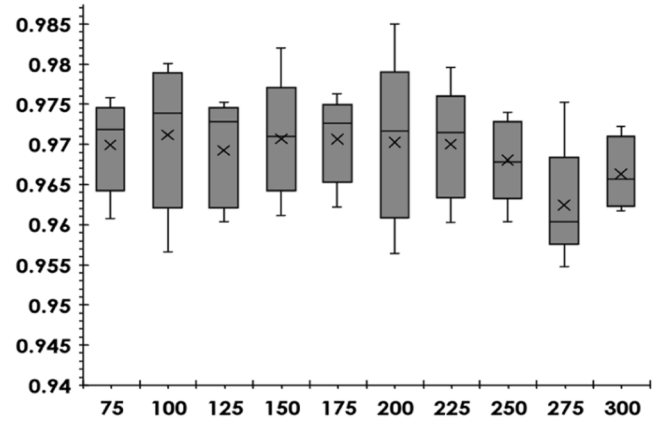


Fig. 8. Scalability analysis for FIDWATCH on BoT-IoT data. X-axis= #clients, and Y-axis=f1-score

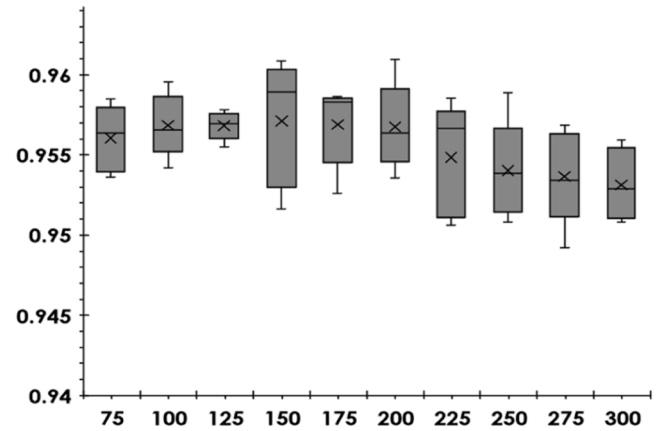


Fig. 9. Scalability analysis for FIDWATCH on ToN-IoT data. X-axis= #clients, and Y-axis=f1-score

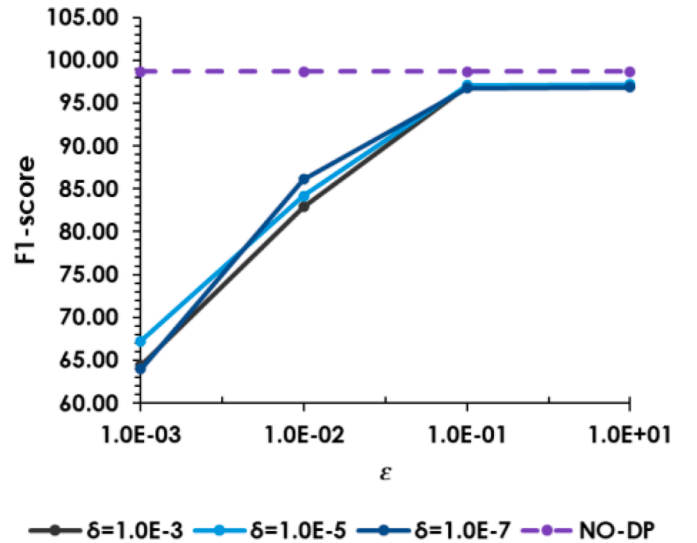


Fig. 10. privacy analysis for FEDWATCH on BoT-IoT data

methods for detecting Windows attacks, millions remain unnoticed. To add insult to injury, DL algorithms are also susceptible to zero-day assaults and the pervasive issue of idea drift. The IoT security threats detection challenge is further complicated by other factors, such as the

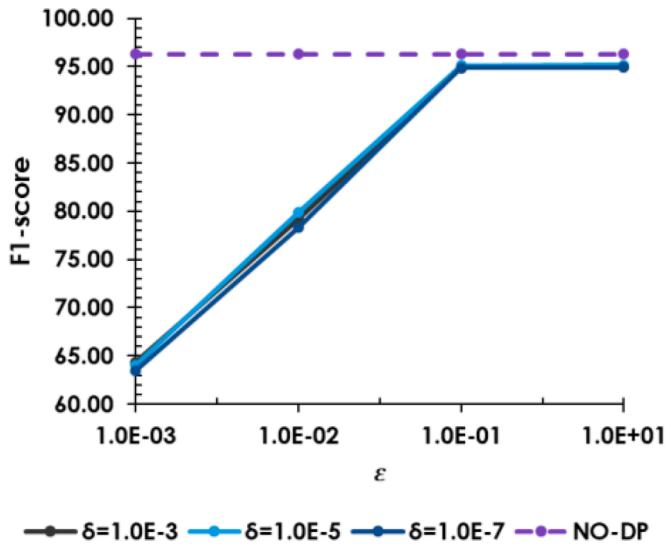


Fig. 11. privacy analysis for FEDWATCH on ToN-IoT data

requirement to deal with a broad range of security threats, a wide range of operating systems, and a large number of CPU designs.

7.1. Data scarcity

Linux security threats recently became the focus of cybersecurity professionals around 2014, making ELF data far less realistic than Windows PE security threats. till now, there has not been any publicly available large benchmarking IoT dataset; nevertheless, the MALNet was just released as an expanded dataset with around 2 million security threats. Suppliers like VirusTotal and

VirusShare is making great strides, but there is still a huge discrepancy between the number of samples accessible for each processor chip and operating system. There is a class imbalance across security threat categories and between security threat instances encoded and converted to various ISAs since it is difficult to build huge data in such complicated contexts with a wide variety of devices, vendors, architects, and operations [35].

7.2. Limited resources

Most IoT devices are designed to run continuously without being powered down despite having extremely limited processing power, data storage space, and other hardware components. Due to their limited hardware resources, they cannot do complex calculations like personal computers based on the AMD/86 technology.

7.3. Absence of standardizations

IoT devices become more susceptible to hacking as the number of connections between them increases exponentially. Managing the information generated by IoT devices is an already complex big data issue. A corrupted system in a network increases the likelihood that other nodes will also be hacked. Since there aren't any universal IoT interoperability standards yet, there are also problems with interoperability across devices from various IoT device vendors. It is also unclear which operating systems the binaries are intended to function in due to the absence of data interoperability. For example, the 'OS/ABI' field in the ELF header is inadequate and underutilized. It's conceivable to use OS/ABI settings that vary from the viable OSes for which the code was created. Another major problem with reverse engineering analyses of firmware is the difficulty of reliably extracting information and meta-data by determining the correct firmware [36]. Developers of IoT

devices are placing minimal value on implementing sufficient security and encryption methods due to pressures to reduce time-to-market and improve features like simplicity and speed. These factors lead to widespread hacking of IoT devices, such as the case involving D-Link routers, when numerous compromised routers were outfitted with back doors, granting security threats authors direct control over the devices used as weaponry in cyber-attacks. Furthermore, most IoT device end-users do not alter the factory-issued identities, leaving their devices open to malicious activities. Embedded systems delivery lacks standards, making it difficult to unpack and decode bespoke formats even if they existed. This is in contrast to conventional software applications, which have established interfaces for the system assets, coding, and file organizations [37].

7.4. Model complexity

The information abstractions used to learn algorithms for cross-architecture IoT security threats detection must be generated utilizing compact cross-architecture sign generation approaches to enhance model tightness and precision, as well as resource effectiveness and power efficacy for localized implementation. Explainability, robustness, and innovative obstinate threat mitigation are all issues that have been identified in previous surveys as potential areas for further study. Language models, computer vision, and graph intelligence techniques were initially offered for learning IoT security threats, and they are the areas from which most extant security threat detection tools on ELF have been adopted. Problems that are unique to ELFs, such as the search for platform-agnostic security threats, have prompted researchers to use methods from other fields of study, such as software flaw discovery. A possible path for future study is to use common ELF properties to facilitate security threat detection in interconnected devices across different architectures [38].

8. Conclusions

This research work presents our efforts to address the limitations of existing intrusion detection systems by developing FIDWATCH as a distributed ML approach for the reliable detection of IoT security threats. FIDWATCH guarantees resource efficiency by presenting a fine-grained KD method while adapting learning capabilities to underlying data scenarios. FIDWATCH addresses the catastrophic forgetting issue through the introduction of an incremental updating strategy to guarantee continuous model updating over time based on perfect alignment between local models and the global model. Proof-of-concept evaluation on benchmark datasets shows that the FIDWATCH is an effective tool for monitoring and detecting security threats in dynamic IoT. The proposed method can overcome the existing SOTA baselines regarding precision, privacy preservation, and compatibility. Finally, we highlight the open research challenges in the field of IoT security to bring the attention of academia toward unexplored areas, promoting further research in this subject matter. The future work includes further refining and enhancing the FIDWATCH to provide a robust, scalable, and versatile solution for securing IoT and Ad Hoc networks against evolving security threats. This can be achieved by extending the FIDWATCH to other Ad Hoc networks and incorporating hybrid models combining federated learning and emerging technologies such as blockchain and edge computing.

Institutional review board statement

The study did not involve humans or animals.

Fund

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Informed consent statement

The study did not involve humans.

CRedit authorship contribution statement

Ibrahim Alrashdi: Writing – original draft, Validation, Investigation, Formal analysis, Conceptualization. **Karam M. Sallam:** Writing – review & editing, Visualization, Validation, Methodology, Data curation, Conceptualization. **Majed Abdullah Alrowaily:** Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Omar Alruwaili:** Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Bilal Arain:** Writing – review & editing, Visualization, Resources, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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